ALAGAPPA UNIVERSITY JOURNAL OF BIOLOGICAL SCIENCES (AUJBS)

Vol 1 No 1 March 2017 ISSN:

Reach at

VI Floor Science Block
Alagappa University
Karaikudi 630 003 Tamil Nadu India
alubiojournal@gmail.com

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Biogenic synthesis of silver nanoparticles from *Cardiospermum halicacabum* decorated with Graphene oxide for enhancing antibacterial ability

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Received: 20.01.2017 / Accepted: 27.02.2017
Published online: 25.03.2017

Abstract Biogenic synthesis is most advantageous synthesis method for the production of active silver nanoparticles (AgNPs) to apply against multi resistant pathogens. However we selected *Cardiospermum halicacabum* a most commonly used medicinal plant for the AgNPs production. The present study faces the green approach in synthesis of AgNPs with Graphene oxide (GO) and compared the activity of both AgNPs & GO-Ag for antibacterial analysis. The synthesised AgNPs and GO decorated AgNPs were characterized individually by ultraviolet-visible Spectroscopy (UV-vis), Fourier Transform Infra-red Spectroscopy (FTIR), X-Ray diffraction (XRD) and High resolution Scanning electron Microscopy (HR-SEM). The results demonstrated that formation of AgNPs in UV-vis showed a surface plasma resonance peak at 440 nm, with diameter of 30 nm and were conveniently deposited on GO sheet. The Antibacterial ability was analysed for both AgNPs & GO decorated AgNPs against *Pseudomonas aeruginosa*, *Klebsiella pneumoniae*, *Staphylococcus aureus* & *Escherichia coli* by Agar well diffusion technique. The results confirm the GO decorated AgNPs nanocomposites excellent activity than the plain AgNPs, which reveals GO-AgNps is a promising tool for wide ranging bio-medical applications.

Key Words AgNPs, Graphene oxide, *Cardiospermum halicacabum*, antibacterial activity, biogenic nanoparticles,

Introduction

The synthesis of potential antibacterial materials is a wide range problem for control growing multi resistant pathogens which may leads to cause severe diseases. Silver plays a vital role in antibacterial applications (Ribeiro et al., 2014; Sivaprakash et al., 2017) and also owing rapid applications in optical, photo-electric, catalytic fields (Sharma et al., 2009). Morover, AgNPs having importance in biomedical, drug delivery, water treatment, antioxidant, antibacterial & anti cancer applications (Boca et al., 2011; Sankar et al., 2013; Shrivastava et al., 2010). Particle size is the major factor strongly related to the antibacterial ability of AgNPs (Wei et al., 2014).These extend a attraction towards new routes for the synthesis of AgNPs (Pugazhenthiran et al., 2009). *Cardiospermum halicacabum* are traditionally used by medicinal practitioners for the treatment of jaundice, asthma, bronchitis, anemia, tuberculosis, urinary bladder disturbances, intestinal infections, diabetes, hepatitis B, Oedema, Nephritis, Oliguria haemorrhoids, pyodermas, earaches, ophthalmias, rheumatism and arthritis (Danso et al., 2013; Shekhawat et al., 2013).

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There are numerous plants utilized for medicinal purposes traditionally due to the presence of alkaloids, flavonoids, glycosides, vitamins, tannins and coumarins (Singh, 2014). The commercial antibiotics are owing less ability against pathogens (Bao et al., 2011; Sahni et al., 2013). So the demand on potential antibacterial agents increased day by day and becoming crucial (Shao et al., 2015). Silver based materials exhibit a potential antimicrobial activity against bacteria (El-Rafie et al., 2013; Gopinath et al., 2013; Shrivastava et al., 2010). AgNPs proves as an nontolerant disinfectant which is having capability to reduce bacterial infections, compared with usage of biocides (Shao et al., 2015). Graphene oxide is well-known monolayers of carbon atom which provides hydroxyl & carboxylic groups at the edges (Wang et al., 2012). Additionally, it consists of functional properties large surface area, low cytotoxicity and water stability. So it having the capability of holding Ag ions and metal ions that can stabilize on them (Han et al., 2013). Physisorption and charge transfer interaction takes place in GO because of interaction between GO and AgNPs in the presence of hydroxyl carboxylic groups (Hui et al., 2014). The proposed theory for AgNPs against microorganism activity is based on electrostatic interaction between the negative charged cell membrane of microbes along with positive charged Ag ions (Shao et al., 2015). However, Ag ions directly affects the bacterial cell which leads to cell death and suppressing DNA replication (Bindhu and Umadevi, 2015; Madhavan et al., 2014). AgNPs activity may be reduced in the case of reduction of their surface area in precipitation, stability and bacterium - nanoparticles interaction (Jain and Pradeep, 2005; Li et al., 2006; Shrivastava et al., 2010). To overcome this problem AgNPs can be bind with suitable material. We found Graphene oxide as an exact material to incorporate with AgNPs and it can form stable Go-Ag. In this present work we synthesized AgNps from *C. halicacabum*, which is decorated with Go and compared for antibacterial application individually. Antibacterial activity was investigated against *Pseudomonas aeruginosa, Klebsiella pneumoniae, Staphylococcus aureus* and *Escherichia coli* for both AgNPs and Go-Ag. The results indicated that Go-Ag has excellent activity than the AgNPs. The results were compared with standard antibiotics.

Materials and methods

Plant collection and processing

C. halicacabum plant samples were collected near science campus of Alagappa University and transferred to laboratory immediately then dried after washing. Leaves were separated and grinded with mixer grinder to get fine powder. 5 g plant leaf powder was boiled with 100 ml of double distilled water for half an hour at 80°C. Final extract was filtered in whatman No.1 filter paper.

Silver nanoparticles synthesis

Aqueous extract (15 ml) was stirred with 50 ml of 1 mm Silver nitrate in 250 ml Erlenmeyer flask for 4 h at room temperature (Krishnaraj et al., 2010). The reaction was terminated after the formation of pale red colour. Aqueous leaf extract and pure $AgNO_3$ was used controls for experimental analysis.

Synthesis of GO decorated silver nanoparticles

AgNPs decorated GO nanocomposite was synthesized as referred by Cheviron *et al* (2014) method with slight modification. 0.2 % of GO was mixed with 20 ml of deionized water and treated under ultrasonication. For

the achievement of 2 mM concentration 10 ml of $AgNO_3$ was slowly added with the above solution. In result brownish colour changes in to gray and attains green colour finally. The obtained Ag-GO was washed several times with deionized water and dried at 100°C for 12h.

Characterization of AgNPs and Ag-GO

The characteristic surface plasma resonance was determined by UV-Vis spectroscopy (Shimadzu, Model UV-1800, Kyoto, Japan) in the ranges between 200-600 nm with 1 nm operation resolutions. FT-IR study was carried in between the range of 400 to 4000cm^{-1}. The crystalline behaviour of AgNPs and Ag-GO were analysed through X-ray diffraction [X'Pert Pro analytical diffractometer, Japan using monochromatic nickel-filtered Cu Ka radiation (k = 0.15405 nm)] and average crystalline size were calculated by Debye scherrer equation ($D=0.9\lambda/\beta \cos\theta$). The clear phases were found from powder diffraction file (PDF) database (ICDD, International Centre for Diffraction Data) and confirmed with X'Pert High Score plus Software. The morphological examination was taken using High resolution Scanning Electron Microscopy (HR-SEM) for both AgNPs and Ag-GO.

Antibacterial assay

The antibacterial assay was performed for both AgNPs and Ag-GO nanocomposite by agar well diffusion method. The activity was investigated against *P. aeruginosa, K. pneumonia, S. aureus* & *E.coli* pathogens. Different volume (25 µl, 50 µl and 75 µl) of AgNPs and Ag-GO were suspended on the Muller Hinton agar plates overlaid with the selected pathogens and incubated overnight at 37°C. Zone of Inhibition was recorded and experiment was analyzed in triplicate to get concordant result. The results were compared with commercial antibiotics (Rifampicin, Streptomycin, Ampicillin, Gentamicin, Neomycin, Bacitracin, Erythromycin and Penicillin).

Results and discussion

Characterization of synthesised AgNPs and GO-Ag

The synthesized AgNPs was analysed through UV-vis spectroscopy and the broad spectrum was observed for peak at the range of 440 nm which unveils the presence of surface plasma resonance of AgNPs (Fig.1). Moreover, pure Silver nitrate solution gets peak at 300 nm, but it's gradually endure in the formation AgNPs synthesis and the hump shows at 435 nm indicates the narrow size distribution along the solution.

Fourier transform infra-red spectra study was performed to find the biomolecules role in the reduction of Ag+ ions and capping of AgNPs synthesized from *C. Halicacabum*. The strong IR bands obtained at 3925.88, 3439.03, 2355.84, 2079.31, 1816.97, 1637.15, 1481.43, 1391.90, 1050.43 & 667.96 cm^{-1} (Fig. 2). The appeared bands at 3439.03 & 2355.84 cm^{-1} corresponds to appearance of –OH stretching & aliphatic –C-H stretching (Zahir and Rahuman, 2012). The data completely reveals the presence of different phosphate and hydroxyl groups. The bands at 1637.15cm^{-1} & 1050.43 cm^{-1} are C=C stretching & -C-O-C stretching modes respectively. Moreover the low bands at the range of 667.96cm^{-1} are representing C-Cl stretching. Therefore, the presence of major components such as of alkaloids, flavonoids, glycosides, vitamins, tannins and coumarins in the *C. halicacabum* plant extract are the exact responsible for the observed reduction of AgNPs.

In the synthesis GO-Ag nanocomposite, Ag ions were reduced in to AgNPs and in the Ag ions fully loaded on the broad surface area of GO sheets. Before formation of graphene sheets, it should be reduced into reduced Graphene oxide (RGO). While heating at 100°C in hot air oven it forms RGO. From the X-ray diffraction studies distinct peaks found at 38.11^{o}, 44.29^{o}, 64.44^{o}, 77.39^{o} and 81.54^{o} were subjected to the (111), (200), (220), (311) and (222) for the formation of cubic AgNPs which perfectly matches to the ICDD card no: 01-087-0597 (Ciobanu et al., 2012) (Fig. 3). From the Fig: 3(a) we easily found the distinct peaks of GO and the reduction of RGO is displayed in Fig: 3(b). The synthesized pure AgNPs *C. halicacabum* was separately displayed in Fig: 3(c). AgNPs embedded with RGO is shown in Fig. 3 (d). High resolution scanning electron microscopy (HR-SEM) was used to visualize the morphology of both AgNPs & GO-Ag. It reveals predominantly spherical shape AgNPs formation the sizes range between 30 to 50 nm (Fig. 4). HRSEM images of AgNPs incorporated with GO nanosheets and displays the agglomeration of AgNPs in GO sheets (Fig. 4). This helps to support the material in nucleation and stabilization process.

Antibacterial activity assay

Four pathogenic strains *P. aeruginosa, K. pneumonia, S. aureus* & *E.coli* were used antibacterial assay. Fig. 5 shows the antibacterial effect of standard antibiotics using the *P. aeruginosa, K. pneumonia, S. aureus* and *E.coli* strains. The applied concentration was varied to 25µl, 50µl & 75µl for both AgNPs and GO-Ag nanocomposite. The activity was observed from the clear zone of inhibition and measured diameters for all strains for both AgNPs and GO-Ag nanocomposite were shown in Fig. 6 & 7 respectively. From the four tested bacteria *P. aeruginosa, S. aureus* has a large zone of inhibition of 14 mm and *E.coli* shows smaller zone of inhibition of 12 mm for 75 µl dosages of AgNPs synthesised from *C. halicacabum.* In case of GO-Ag the maximum zone was observed for *P. aeruginosa* (18 mm) and *K. pneumonia* shows small zone of inhibition of 15 mm at 75 µl dosage. There was a significant difference in zone diameter between the pure AgNPs and GO-Ag. It was due to the bacteria susceptibility to the prepared AgNPs and GO-Ag. Moreover AgNPs has capability of oxidative stress in bacterial cell walls (De Faria et al., 2014), the potential ability is increased due to the combination of GO-Ag, which leads to suppress the proteins and enzymes, makes damage in DNA replication (Markowska et al., 2013). The present study clearly shows the excellent antibacterial ability of AgNPs and GO-Ag against both gram positive and negative bacteria. Notably, GO-Ag nanocomposite has all beneficial capability for excellent antibacterial agent in biomedical applications.

Conclusion

In the present study we prepared AgNPs from environmental friendly, simple method by using *C. halicacabum* leaf extract, and it's conveniently decorated on GO sheets. GO has monodispersed size AgNPs, were agglomerated with GO sheets and forms GO-Ag nanocomposite. The synthesized AgNPs and GO-Ag were used for antibacterial ability analysis against *P. aeruginosa, K. pneumonia, S. aureus* and *E.coli*. GO-Ag proves as an effective antibacterial activity than the pure AgNPs. We could conclude that Go-Ag

composite can be used as a promising tool for biomedical applications.

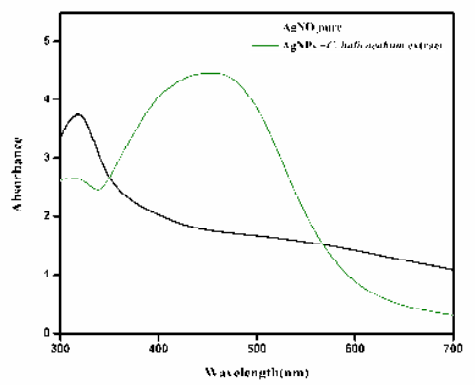

Fig. 1: UV-V is spectra of synthesized AgNPs using *C. halicacabum* extract

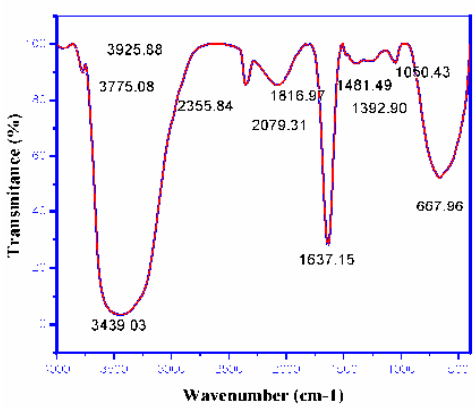

Fig. 2: FT-IR spectra of synthesized AgNPs using *C. halicacabum* extract

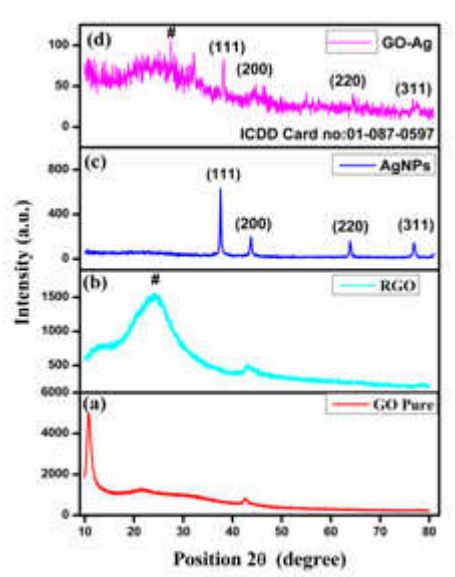

Fig. 3: XRD pattern of Graphene oxide (a), reduced Graphene oxide (b), synthesized AgNPs using *C. halicacabum* extract (c), GO decorated Ag (d)

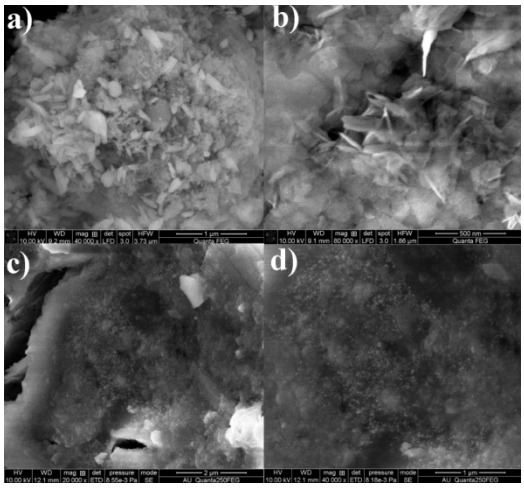

Fig. 4: HR-SEM images of pure AgNPs (a), (b) and GO-Ag (c), (d)

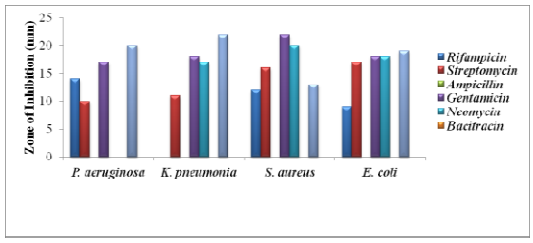

Fig. 5: Antibacterial activity for various antibiotic susceptibility discs

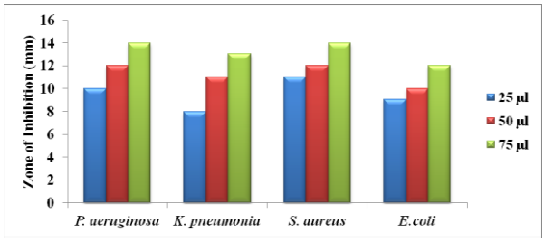

Fig. 6: Antibacterial analysis of AgNPs

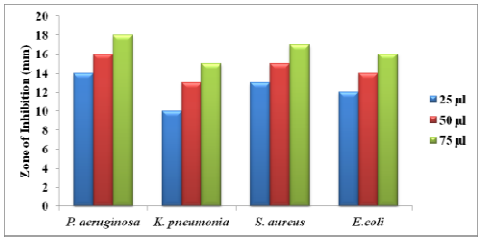

Fig. 7: Antibacterial analysis of GO-Ag

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